Analysis of Zoning Utilization Ratios in Newly Completed NYC Residential Buildings (2012–2025)

Jeremy Gomez

Project Background – The Housing Crisis

Context:

- NYC vacancy rate at historic low: 1.4% (2025)
- Median Manhattan rent: $4,750 (2025)

Driver:

- High costs (regulatory, land, construction) restrict new supply

Utility:

- Economic Utility: Maximize rentable/sellable area to offset high land costs
- Social Utility: Add high-quality homes to reduce demand pressure

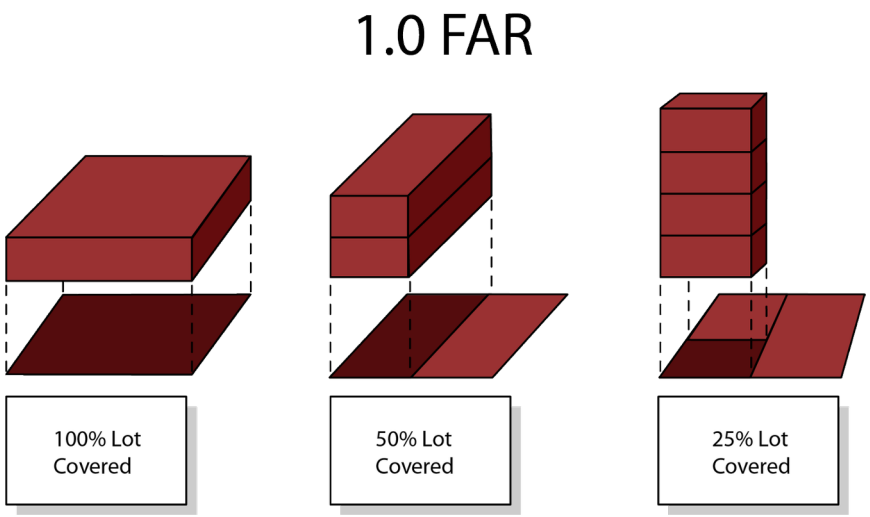


What is FAR (Floor Area Ratio)?

- **Total Building Floor Area** divided by **Lot Area**
- **Example:** FAR = 1.0**:** A 1-story building covering 100% of the lot or a 2-story building covering 50%

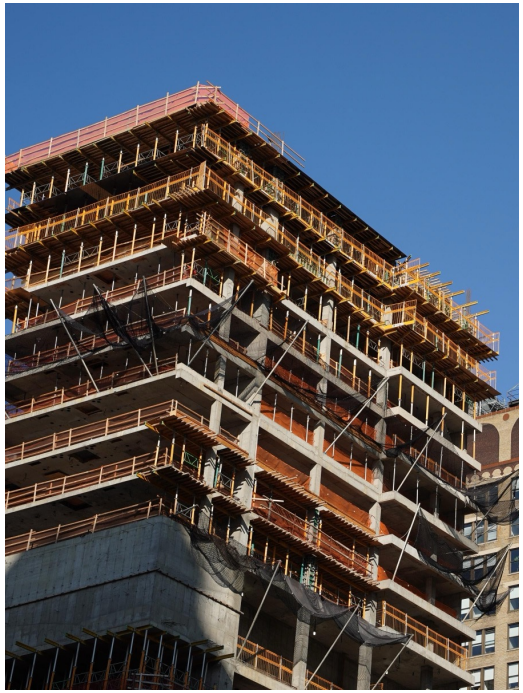
Why it Matters:

- FAR is the primary tool regulators use to control density
- It defines the limit for how much housing can be built on a specific plot



Project Goals

- Determine Predictors: Use Logistic Regression to quantify the impact of variables (Land Value, Zoning, etc.)
- Ensure Robustness: Use Non-Parametric Bootstrapping to validate findings
- Visualize Trends: Use GAM (Generalized Additive Models) to map zoning utilization trends over time against industry shocks (COVID, ZQA, 421-a)



Data Sources

My data originates from four datasets from NYC Open Data

- **DOB Certificate of Occupancy (2012–2021)**
- **DOB NOW (2021–2025**
- **DOB JOB Filings:** Used to filter for residential units
- **PLUTO:** Source for "Allowable FAR" and "Built FAR"

The “Final Dataset” contains 15 variables and over 200k observations

Variable Name	Outcomes	Converted Values	Type
Job_Number	Unique identifiers	Original Values (Character string)	Categorical (Nominal)
Issue_Date	Date values (YYYY-MM-DD)	Original Values	Date / Time
FAR_Utilization	Continuous numeric ratio	Calculated (BuiltFAR / AllowableFAR)	Numerical (Continuous)
Maxed_Out	0, 1	1 (Yes, Ratio > 0.9), 0 (No)	Categorical (Binary)
BuiltFAR	Numeric values	Original Values (Cleaned to numeric)	Numerical (Continuous)
AllowableFAR	Numeric values	Original Values (Cleaned to numeric)	Numerical (Continuous)
Borough	Boroughs: Bronx, Brooklyn, Manhattan, Queens, Staten Island	Original Values (Title Cased)	Categorical (Nominal)
Zoning_Category	R, C, M	Factor levels (First letter of zone)	Categorical (Nominal)
Is_Landmark	No, Yes	Factor levels ("No", "Yes")	Categorical (Binary)
Is_Irregular	false, true	Factor levels ("false", "true")	Categorical (Binary)
LotFront	Numeric values (Linear feet)	Original Values	Numerical (Continuous)
Land_Value_Per_SqFt	Numeric values (USD $)	Calculated (AssessLand / LotSize)	Numerical (Continuous)
Job_Type	New Building, Alteration 1	Original Values	Categorical (Nominal)
Dwelling_Units	Numeric count (Integers ≥ 1)	Original Values	Numerical (Discrete)
LotSize	Numeric values (Square feet)	Original Values	Numerical (Continuous)

Exploratory Data Analysis: Correlation Matrix

	FAR_Utilization	Land_Value	Lot_Frontage	Lot_Size	Dwelling_Units
FAR_Utilization	1	0.2595	0.0012	-0.0917	0.3632
Land_Value	0.2595	1	0.0107	-0.0448	0.4288
Lot_Frontage	0.0012	0.0107	1	0.8747	0.2281
Lot_Size	-0.0917	-0.0448	0.8747	1	0.0232
Dwelling_Units	0.3632	0.4288	0.2281	0.0232	1

Correlation Matrix Findings:

- Land Value vs. Utilization (r = 0.26): Weak positive correlation
- Lot Size vs. Utilization (r = -0.09): Negligible negative correlation
- Collinearity between Lot Frontage and Lot Size

Pearson Chi-Squared Test Results

Chi-Squared Test Results:

- Strongest Predictors:
  - Borough (X2 = **24,157**): Borough is the Strongest Categorical Predictor
  - Zoning Category (X2 = **9,094**): Commercial/Mixed vs. Manufacturing zone is a significant predictor
  - Lot Shape (X2 = **5,901**): Irregular lots behave differently than regular rectangular lots

- Weakest Predictor:
  - Landmark Status (X2 = **211**): Historic status is a constraint, but its predictive power is minor

Variable	X-squared Statistic	Degrees of Freedom (df)	P-Value
Borough	24,157	4	< 0.001
Zoning Category	9,094.40	2	< 0.001
Irregular Lot	5,901.70	1	< 0.001
Landmark Status	211.5	1	< 0.001

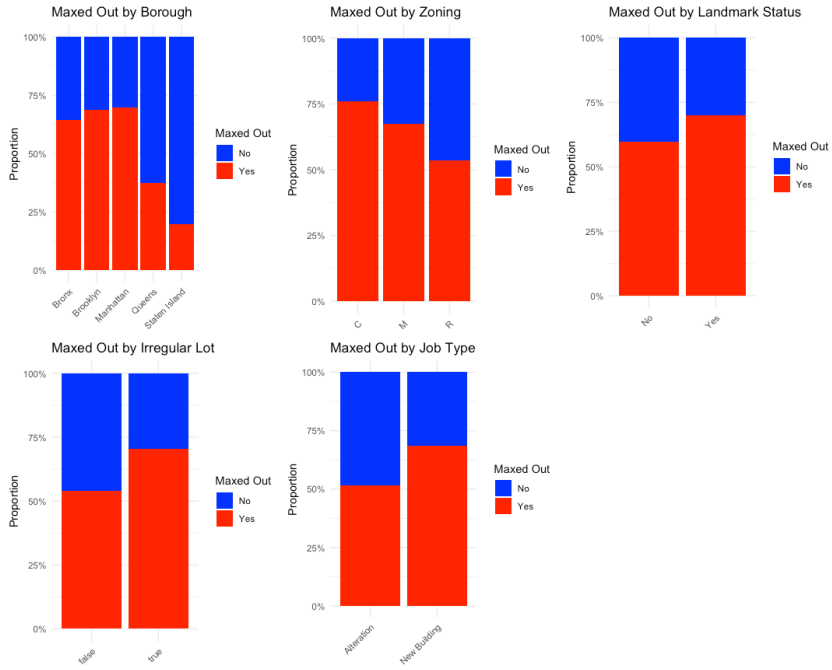
Visual Analysis of Categorical Associations

High Utilization Groups ("Maxed Out"):

- Borough**:** Manhattan & Brooklyn
- Job Type**:** New Buildings
- Lot Shape**:** Irregular Lots (Corner lots maximize light/air exposure).

Low Utilization Groups:

- Borough**:** Staten Island & Queens
- Zoning**:** Manufacturing (M) districts lag behind Residential/Commercial.
- Landmark Status shows a weaker association than Borough or Job Type, suggesting historic status is a less dominant constraint than commonly believed.



Logistic Regression: Model Performance

Model Performance:

Null Deviance**:** 315,086 > Residual Deviance**:** 266,330

Non-Significant Factors (p > 0.05):

- Landmark Status: (p=0.91) – Statistically useless in this model
- Borough (Manhattan): (p=0.052) – Insignificant compared to the baseline (The Bronx)

```
glm(formula = Maxed_Out ~ Borough + Zoning_Category + Is_Landmark +
    Is_Irregular + Job_Type + LotSize + Land_Value_Per_SqFt,
    family = "binomial", data = final_dataset)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)           -2.046e-01  2.305e-02  -8.876   <2e-16 ***
BoroughBrooklyn        4.365e-01  1.823e-02  23.943   <2e-16 ***
BoroughManhattan       3.937e-02  2.031e-02   1.939   0.0525 .
BoroughQueens         -9.506e-01  1.885e-02 -50.434   <2e-16 ***
BoroughStaten Island -1.951e+00  2.731e-02 -71.450   <2e-16 ***
Zoning_CategoryM      -1.959e-01  2.150e-02  -9.110   <2e-16 ***
Zoning_CategoryR      -2.423e-01  1.465e-02 -16.542   <2e-16 ***
Is_LandmarkYes        -3.750e-03  3.362e-02  -0.112   0.9112
Is_Irregulartrue       5.710e-01  1.046e-02  54.614   <2e-16 ***
Job_TypeNew Building   1.075e+00  1.034e-02 103.949   <2e-16 ***
LotSize               -1.051e-07  6.284e-09 -16.729   <2e-16 ***
Land_Value_Per_SqFt    2.913e-03  5.169e-05  56.352   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 315086  on 233989  degrees of freedom
Residual deviance: 266330  on 233978  degrees of freedom
AIC: 266354

Number of Fisher Scoring iterations: 5
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```

Logistic Regression: Results

The "Max Out" Positive Impact:

- Job Type (**New Building**): Log Coef +1.075 (Odds Ratio: **2.93x**)

The "Max Out" Negative Impact:

- **Staten Island:** Log Coef -1.951 (Odds Ratio: **0.15x**)
- **Queens:** Log Coef -0.951 (Odds Ratio: **0.39x**)

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```
exp(coef(logit_model))
      (Intercept)      BoroughBrooklyn    BoroughManhattan       BoroughQueens BoroughStaten Island
        0.8149963           1.5472745          1.0401555           0.3864922           0.1420707
  Zoning_CategoryM    Zoning_CategoryR      Is_LandmarkYes    Is_Irregulartrue Job_TypeNew Building
        0.8221162           0.7848204          0.9962565           1.7701238           2.9306611
         LotSize Land_Value_Per_SqFt
        0.9999999           1.0029171
```

Bootstrapping of Logistic Regression Coefficients

Bootstrap ID	Variable Name	Original Coeff	Std. Error	95% CI (Lower)	95% CI (Upper)
t1*	(Intercept)	-0.2046	0.0222	-0.2481	-0.1611
t2*	Borough: Brooklyn	0.4365	0.0186	0.4	0.473
t3*	Borough: Manhattan	0.0394	0.0213	-0.0023	0.0811
t4*	Borough: Queens	-0.9506	0.0194	-0.9886	-0.9126
t5*	Borough: Staten Is.	-1.9514	0.0247	-1.9998	-1.903
t6*	Zoning: Manufacturing	-0.1959	0.0247	-0.2443	-0.1475
t7*	Zoning: Residential	-0.2423	0.0141	-0.2699	-0.2147
t8*	Is_Landmark	-0.0038	0.0363	-0.0749	0.0673
t9*	Is_Irregular	0.571	0.01	0.5514	0.5906
t10*	Job_Type: New Bldg	1.0752	0.0112	1.0532	1.0972
t11*	Lot Size	-1.05E-07	4.80E-09	-1.14E-07	-9.60E-08
t12*	Land Value Per SqFt	0.0029	0.0001	0.0028	0.003

Methodology:

- Non-parametric bootstrap (100 resamples)
- Generated 95% Confidence Intervals for all coefficients

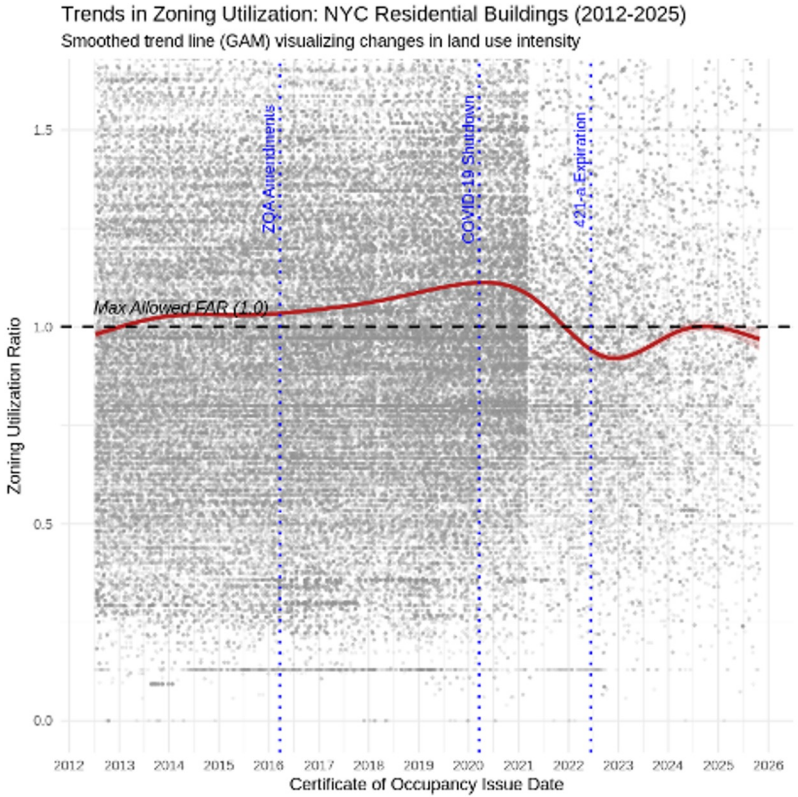
Temporal Analysis: General Additive Model

Why GAM (General Additive Model)?

- Allows for non-linear trend visualization
- Efficient for large datasets (200k+ observations) where LOESS failed due to compute restrictions

Trend Line Patterns

- **2012–2019: Post-Recession & ZQA Act** steady growth in density
- **2020–2022: COVID-19 Pandemic** causes sharp decline below the 1.0 baseline.
- **2022–2025: 421-a Expiration** creates a "Rush-and-Crash" volatility.



Final Analysis

- Land value influences zoning utilization, but regulatory & physical constraints matter more
- Existing buildings are structurally or legally difficult to expand, leaving density on the table
- Staten Island and Queens are "under-building," regardless of allowable FAR



Recommendations for City Policy Makers

1. **Unlock "Alteration" Density:**
    - Make it easier to demolish obsolete structures.
    - New Buildings are 3x more likely to utilize greater than 90% of the legally allowable zoning

2. **Incentivize the Outer Boroughs:**
    - Introduce density bonuses/parking reductions for Staten Island & Queens
    - Developers in these areas are voluntarily under-building. The market incentives

3. **Restore Tax Incentives (485-x):**
    - Ensure the successor to 421-a is robust
    - The data shows zoning utilization drops when tax incentives expire